

## Article

# Image Processing in $L^1$ -Norm-Based Discrete Cartesian and Polar Coordinates

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**Abstract:** This paper proposes a radial image processing method performed in an  $L^1$ -norm-based discrete polar coordinate system. For this purpose, we address the problem that polar coordinates based on the  $L^2$ -norm cannot exist in discrete systems and then develop a method for converting Cartesian coordinates to  $L^1$ -norm-based discrete polar coordinates. The proposed method greatly reduces the directional variance occurring in the Cartesian coordinate system and so processes radial directional images along the directions of the local image signal flows. To verify the usages of the proposed method, it was applied to the stabilization of mass-type breast cancer images, a segmentation of extremely deformable objects such as biomedical objects. In all cases, the proposed method produced superior results compared to the processing in the Cartesian coordinate systems. The proposed method is useful for processing or analyzing diffusing and deformable images such as bio-cell and smoke images.

**Keywords:** discrete polar coordinate; discrete Cartesian coordinate; radial image processing; biomedical image processing



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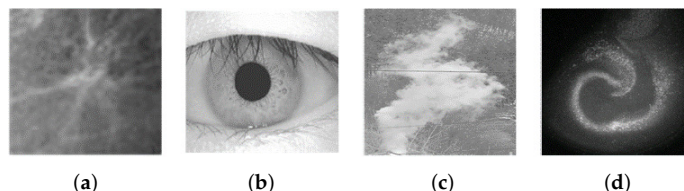
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## 1. Introduction

Many images have radial characteristics such that their image signals propagate from their centers [1]. Figure 1 shows representative examples of such images. Biomedical image signals such as mass lesions in mammographic images tend to diffuse radially [2,3]. In eye images, the iris and pupil areas are circularly separated at some distance from the center of the eyes [4]. Since such image signals are dominated by radial directions, processing the images in the Cartesian coordinate system, i.e., rectangular coordinate system, inherently causes the directional variant problem. It is more appropriate to process such images radially in a polar coordinate system [5,6].



**Figure 1.** Examples of radial image signals. (a) Mass-type breast cancer, (b) iris, (c) plume, (d) hippocampus.

Because pixels in the discrete image domain are regularly arranged squares, the set of pixels forms a  $Z^2$  lattice structure. The coordinates of pixels are the centers of lattice areas. The pixels of images digitally captured form a  $Z^2$  lattice grid in Cartesian coordinate systems, where the pixel coordinates are the centers of the lattices [7]. To process an

image radially in this lattice structure, the pixel locations between the discrete Cartesian coordinates and the discrete polar coordinate must be perfectly convertible. Therefore, the pixel locations between two coordinates must be in a one-to-one mapping, implying that a coordinate pointing to a pixel must be mapped to one other coordinate.

Conventional radial image processing methods adopt the  $L^2$ -norm distance measure [5]. These methods presume the image domain to be continuous and map the numerically closest locations to convert the pixel locations to the locations in other coordinate systems. Since the numerically closest locations do not match the geometrical locations, the methods inherently induce the coordinate transform mismatches that are larger at further distances from the center. To compensate for these mismatches, conventional methods require additional interpolations [5]. Because the mismatches are proportional to the increase in the distance from the center, the interpolation accuracies are reduced more at the pixel further from the center. Consequently, these methods attenuate the radial directional property of a radial image signal as the distance from the center increases [1].

In order to overcome conventional radial image processing methods, a radial processing method based on the  $L^1$  distance measure was developed in [8]. The proposed method exactly converts the pixel locations between the discrete Cartesian coordinate system and the discrete polar coordinate system, thus fundamentally eliminating the coordinate transform mismatches. However, the method in [8] cannot directly transform Cartesian images to polar images. In [8], the converted polar images were the lists of indices mapping Cartesian coordinates. Thus, the contents of the images in the polar coordinate were not recognized because they appeared as an array of indices. Therefore, the method in [8] can be applied to classic image processing techniques but not to advanced techniques accompanied by learning. This study developed a method that can directly convert Cartesian images to polar images. Therefore, unlike conventional methods, the developed method is applicable not only to any classical image processing technique but also to image processing methods that require learning.

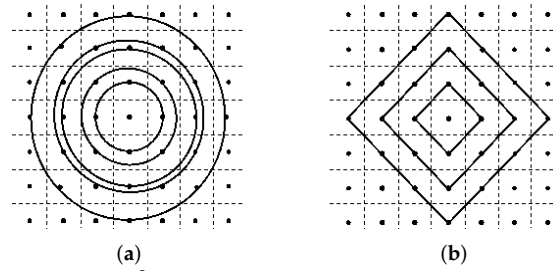
We verified the effectiveness of the proposed method by providing exemplary applications. The applications include smoothing the mass image signal in a mammographic image and segmenting extremely deformable objects with Mask-RCNN [9]. In all cases, the proposed method outperformed the methods in the Cartesian coordinate system and did not require any additional processing for compensation for coordinate transform mismatches. The essence of the proposed method is that the method can easily track the image signal's local directions by greatly reducing the directional variance and so process the image signal adaptively to the signal's local directions [8].

This paper is organized as follows. In Section 2, we verify the infeasibility of the  $L^2$ -norm-based discrete polar coordinates and explain how the  $L^1$ -norm-based discrete polar coordinate is constructed. Section 3 develops an  $L^1$ -norm-based coordinate transform algorithm. The radial image proceeding method using the  $L^1$ -norm-based coordinate transform algorithm is developed in Section 4. Section 5 presents several applications of the developed radial image processing method. Finally, Section 6 concludes the paper and discusses the further studies for improving the performance of the proposed method.

## 2. $L^1$ - and $L^2$ -Norm-Based Discrete Image Domains

Image pixels in a digital domain are on a regularly spaced grid, and the set of pixels is modeled as the  $Z^2$  lattice structure [7]. Figure 2 illustrates the lattice structure for presenting a discrete image domain. The pixel resolution is the area of the lattices. The coordinates of the pixels are the centers of squares, i.e., lattice points, where black dots indicate the centers of pixel areas which are the coordinates of the pixels [10].

Using the  $L^2$ -norm distance, the set of all pixels equidistant from the origin forms a circle. Alternatively, using the  $L^1$ -norm distance, the set of all pixels equidistant from the origin forms a pyramid. Figure 2 illustrates the distribution of pixels on the circles using the  $L^2$ -norm distance and on the pyramids using the  $L^1$ -norm distance.



**Figure 2.**  $Z^2$ -lattice structures of pixel areas. Distribution of pixels belonging to (a) circles using  $L^2$ -norm distance and (b) pyramids using  $L^1$ -norm distance.

In a discrete Cartesian coordinate system,  $L^2$ -norm-based equations that convert pixel positions from  $(m, n)$  to  $(r, \theta)$  in a discrete polar coordinate system are as follows:

$$r = \sqrt{m^2 + n^2}, \quad \theta = \tan^{-1}\left(\frac{n}{m}\right), \quad m, n \in \mathbb{Z} \quad (1)$$

Conversely, to convert  $L^2$ -norm-based  $(r, \theta)$  in a polar coordinate system to  $(m, n)$  in a discrete Cartesian coordinate system, we must calculate continuous values using trigonometric functions. However, the pixel position as a discrete coordinate of a digital image domain should be expressed in integers. Therefore, the continuous values should be approximated to integer values. Thus, equations that convert pixel positions from  $(r, \theta)$  in a discrete polar coordinate system to  $(m, n)$  in a discrete Cartesian coordinate system are as follows:

$$m = \partial(r \cos(\theta)), \quad n = \partial(r \sin(\theta)) \quad (2)$$

where  $\partial(\alpha)$  is the rounding operation that rounds the closest integer to  $\alpha$ .

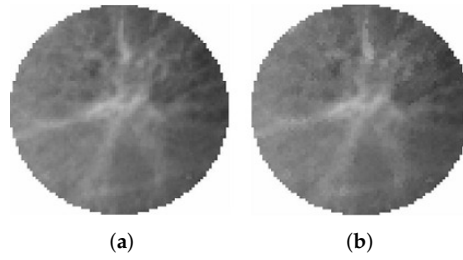
Figure 2a illustrates the distribution of pixels on the circles. The radii of the circles where the pixels are located increase irregularly (for example,  $r = 1, \sqrt{2}, 2, \sqrt{5}, 2\sqrt{2}$ , etc.). The number of pixels located on each circle also increases irregularly corresponding to the increase in radius (for example,  $N = 4, 4, 4, 8, 4$ , etc.). Thus, if  $L^2$ -norm is used, it is impossible to find a one-to-one mapping between the coordinate systems, and, therefore, it is not possible to construct a discrete polar coordinate system mathematically. Thus, using  $L^2$ -norm, it is impossible to determine the discrete Cartesian coordinate  $(m, n)$  which corresponds to a polar coordinate  $(r, \theta)$ . For example, the discrete Cartesian coordinate closest to a point  $1 \cdot e^{j(24/100)\pi}$ , whose actual radius is 1, is  $(1, 1)$ , of which the radius is not  $r = 1$  but  $r = \sqrt{2}$ .

Figure 3 shows an example of image reconstruction using  $L^2$ -norm, which converts an image to the polar coordinate system and then converts it back to a Cartesian coordinate system. The reconstructed image is not the same as the original one as shown in Figure 3. The PSNR value for the reconstructed image in the case of mammogram images is around 35 dB. Therefore, to convert an image from a polar coordinate system to a discrete Cartesian coordinate system, the  $L^2$ -norm-based radial processing requires additional processing such as interpolation or stretching to compensate for the coordinate transform mismatch.

On the contrary, in a discrete Cartesian coordinate system,  $L^1$ -norm-based equations that convert pixel positions from  $(m, n)$  to  $(r, \theta)$  in a discrete polar coordinate system are as follows:

$$r = |m| + |n|, \quad \theta = \tan^{-1}\left(\frac{n}{m}\right), \quad m, n \in \mathbb{Z} \quad (3)$$

As shown in Figure 2b, the radii of the pyramids where the pixels are located increase regularly ( $r = 0, 1, 2, 3, 4$ , etc.). The total number of pixels located on each pyramid also increases regularly ( $N = 1, 4, 8, 12, 16, \dots = 4r$ ) [10,11]. Therefore, if  $L^1$ -norm is used, it is possible to construct a discrete polar coordinate system. The pixel locations can also be completely converted between the discrete Cartesian and the discrete polar coordinate systems.



**Figure 3.** An example of image reconstruction using  $L^2$ -norm: (a) Original image and (b) image converted to polar coordinates and then converted back to Cartesian coordinates. The PSNR between two images is about 35 dB.

In the following section, we derive equations that convert pixel positions from  $(r, \theta)$  in the  $L^1$ -norm-based discrete polar coordinate system to  $(m, n)$  in the discrete Cartesian coordinate system.

### 3. $L^1$ -Norm-Based Discrete Coordinates

In this section, we develop a method for transforming the coordinates between the discrete Cartesian coordinate system and the  $L^1$ -norm-based discrete polar coordinate system.

Let  $(r, \theta)$  be the location in the discrete polar coordinate system corresponding to  $(m, n)$  in the discrete Cartesian coordinate system. Figure 4 shows the relation between  $(r, \theta)$  and  $(m, n)$  in the first quadrant. Due to the symmetrical property of the coordinate systems,  $(r, \text{mod}(\theta, 90^\circ))$  is in the first quadrant with respect to  $(r, \theta)$  in any quadrant, where  $\text{mod}(\alpha, \beta)$  is a modulo operation, which is equal to the remainder when  $\alpha$  is divided by  $\beta$ . Therefore, the symmetric location  $(|m|, |n|)$  in the first quadrant of the discrete Cartesian coordinate becomes  $(r, \text{mod}(\theta, 90^\circ))$  of the discrete polar coordinate. This implies that if we derive  $(|m|, |n|)$  in the first quadrant with respect to  $(r, \text{mod}(\theta, 90^\circ))$ , we can determine the  $(m, n)$  in any quadrant.

As seen in Figure 4, it is reasonable that a pixel at  $(r, \text{mod}(\theta, 90^\circ))$  can be converted to the geometrically closest pixel point  $(|m|, |n|)$  which is on the pyramid of radius  $r$ . Both  $(|m|, |n|)$  and  $(r, \text{mod}(\theta, 90^\circ))$  are on the same pyramid. From the nearest neighbor theory in Voronoi cells [7], the angle is close to  $\text{mod}(\theta, 90^\circ)$  compared to any other points on the radius  $r$ . Thus,  $(|m|, |n|)$  satisfies

$$\left| \tan^{-1} \left( \frac{|n|}{|m|} \right) - \text{mod}(\theta, 90^\circ) \right| < \left| \tan^{-1} \left( \frac{|n'|}{|m'|} \right) - \text{mod}(\theta, 90^\circ) \right| \quad (4)$$

where  $(m, n) \neq (m', n')$ , and  $(m', n')$  is also a lattice point on the pyramid as  $(m, n)$ , which satisfies  $|m'| + |n'| = r$ .

Since the tangent function increases monotonically at the first quadrant, Equation (4) is equivalent to the following

$$\left| \frac{|n|}{|m|} - \tan(\text{mod}(\theta, 90^\circ)) \right| < \left| \frac{|n'|}{|m'|} - \tan(\text{mod}(\theta, 90^\circ)) \right| \quad (5)$$

From  $\tan(\text{mod}(\theta, 90^\circ)) = |\tan \theta|$ ,  $|m| + |n| = r$  and  $|m'| + |n'| = r$ ; therefore, Equation (5) becomes

$$\left| \frac{|n|}{r - |n|} - |\tan \theta| \right| < \left| \frac{|n'|}{r - |n'|} - |\tan \theta| \right| \quad (6)$$

From (6),  $|n|$  should satisfy the following:

$$|n| = \arg \min_{0 \leq |n| \leq r} \left\{ \left| \frac{|n|}{r - |n|} - |\tan \theta| \right| \right\} \quad (7)$$

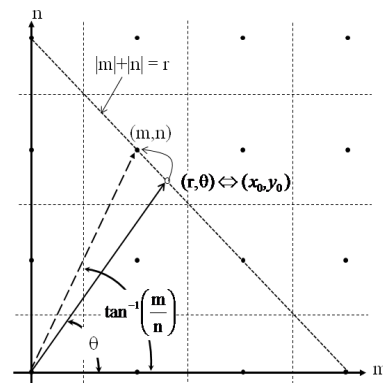
In order to find  $|n|$  in (7), we define a function  $\mathcal{D}(y)$  of a real variable  $y$  such as the following:

$$\mathcal{D}(y) = \frac{y}{r - y} - |\tan \theta| \quad (8)$$

where  $0 \leq y < r$ . The  $\mathcal{D}(y)$  is a monotonously increasing function and  $\mathcal{D}(0) < 0$ ,  $\mathcal{D}(r - \epsilon) > 0$  ( $\epsilon > 0$ ). Then, from the intermediate value theorem [12], there must exist only one solution  $y_0$  which satisfies  $\mathcal{D}(y_0) = 0$ . By solving  $\mathcal{D}(y_0) = 0$ ,  $y_0$  is as follows:

$$y_0 = \frac{r \cdot |\tan \theta|}{1 + |\tan \theta|} \quad (9)$$

The continuous Cartesian coordinate corresponding to  $(r, \theta)$  is  $(x_0, y_0)$ , where  $r = |x_0| + |y_0|$ .



**Figure 4.** A lattice mapping from a continuous Cartesian coordinate to an  $L^1$ -norm-based discrete polar coordinate.

As shown in Figure 4, if  $x_0 = r - y_0$ , then  $(x_0, y_0)$  is the real coordinate with respect to  $(r, \text{mod}(\theta, 90^\circ))$ . Therefore, the discrete Cartesian coordinate  $(|m|, |n|)$  corresponding to the polar coordinate  $(r, \theta)$  is the integer coordinate closest to  $(x_0, y_0)$  on the radius  $r$ . Thus, Equation (7) becomes

$$\begin{aligned} |n| &= \arg \min_{0 \leq |n| \leq r} \left\{ \left| \frac{|n|}{r - |n|} - |\tan \theta| \right| \right\} \\ &= \partial(y_0) = \partial\left(\frac{r \cdot |\tan \theta|}{1 + |\tan \theta|}\right) \end{aligned} \quad (10)$$

where  $\partial(\alpha)$  is a function to find the closest integer from real number  $\alpha$ .

Using  $r = |m| + |n|$  and the symmetrical property of the coordinate systems, the coordinate position  $(m, n)$  at any quadrant can be determined from the symmetrical position  $(|m|, |n|)$  at the first quadrant as follows:

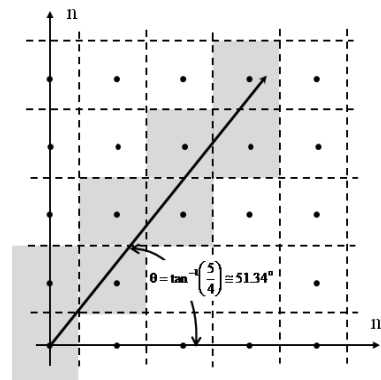
$$(m, n) = \begin{cases} (r - |n|, |n|) & \text{for } 0^\circ \leq \theta < 90^\circ \\ (-r + |n|, |n|) & \text{for } 90^\circ \leq \theta < 180^\circ \\ (-r + |n|, -|n|) & \text{for } 180^\circ \leq \theta < 270^\circ \\ (r - |n|, -|n|) & \text{for } 270^\circ \leq \theta < 360^\circ \end{cases} \quad (11)$$

From (11), we can find the discrete Cartesian coordinate position  $(m, n)$  corresponding to  $(r, \theta)$  by determining  $(|m|, |n|)$  from (10).

As discussed above, the discrete Cartesian coordinate  $(m, n)$  corresponding to the discrete distance  $r (= 0, 1, 2, 3, 4, 5, \dots)$  at a certain angle  $\theta$  can be determined from (10) and (11). Figure 5 shows the lattice points on the line drawn at  $\theta = \tan^{-1}(5/4) (\simeq 51.34^\circ)$  with distance  $r$  from 0 to 7. In the case of  $\theta = 51.34^\circ$ , we found the following relation:

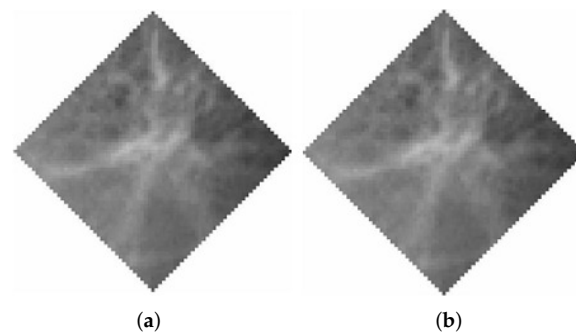
$$\begin{aligned} & \{(r, \theta = 51.34^\circ) | (0, \theta), (1, \theta), (2, \theta), (3, \theta), (4, \theta), (5, \theta)\} \\ &= \{(m, n) | (0, 0), (0, 1), (1, 1), (1, 2), (2, 2), (2, 3)\} \end{aligned}$$

As shown in the above example, the only discrete Cartesian coordinate  $(m, n)$  corresponding to each discrete distance  $r$  at a certain angle  $\theta$  can be determined. There exists a complete one-to-one mapping between the discrete Cartesian coordinate  $(m, n)$  and the polar coordinate  $(r, \theta)$ .



**Figure 5.** Lattice points on the line drawn at  $\theta = \tan^{-1}(5/4) (\simeq 51.34^\circ)$  with distance  $r$  from 0 to 7. Grey areas are the lattices that each lattice point covers.

Figure 6 shows an example of image reconstruction using the  $L^1$ -norm. We convert an image to the polar coordinate system using (7) and (11) and then convert it back to the Cartesian coordinate. To verify the robustness of the developed method, Gaussian noise was added to the extent that the image quality did not change. The PSNR value of the reconstructed image was infinity, which means the quality of the reconstructed image was exactly the same as the original one as shown in Figure 6. This experimentally verifies that there exists a complete one-to-one mapping between the discrete Cartesian coordinate system and the polar coordinate system.



**Figure 6.** An example of image reconstruction using  $L^1$ -norm: (a) original image and (b) image converted to the polar coordinate system and then converted back to the Cartesian coordinate system. The two images are exactly the same.

#### 4. Radial Image Processing in Discrete Polar Coordinate

In this section, we develop a radial image processing method that functions in the  $L^1$ -norm discrete polar coordinate system. The proposed method reduces the directional variance by representing radial directional image signals in the discrete polar coordinate

system. This makes it possible to process the radial image signal in the direction of the signal's local circular flows. Therefore, the proposed method preserves the directional features of the image regardless of the distance from the center.

Using (10) and (11), the pixel locations at a given angle in the Cartesian coordinate system are transformed to the corresponding location in the polar coordinate system. This procedure is performed at all angles to complete the coordinate transformation. Then, the desired processing is applied to the image plane represented in the polar coordinate system. The images processed in polar coordinates are again converted back to the Cartesian coordinate system. It should be noted that pixel locations at a specific angle can be arranged only in  $L^1$ -norm-based discrete polar coordinates. Consequently, the pixels at each line of a radial axis in the polar coordinate are the ones on the pyramid in the Cartesian coordinate.

Algorithm 1 describes the detailed implementation of the proposed method. The legitimate angles are determined by the locations of pixels on the pyramid enclosing target images. Let  $R_{max}$  be the radius of the pyramid enclosing the images. Because there are  $4 \cdot r$  lattice points on the pyramid with radius  $r$ , the number of legitimate angles is  $4 \cdot R_{max}$  [7,11]. Legitimate angles are arranged in a counter-clockwise direction. Although the angle intervals in the Cartesian coordinate system are irregular, the angle values are directly mapped to the locations of the discrete polar coordinate. Therefore, the polar coordinate is completely discrete, and the horizontal and vertical directions are the angle and radial directions, respectively. Figure 7 illustrates the scheme of the proposed method.

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**Algorithm 1:** Implementation of the proposed coordinate conversion method.

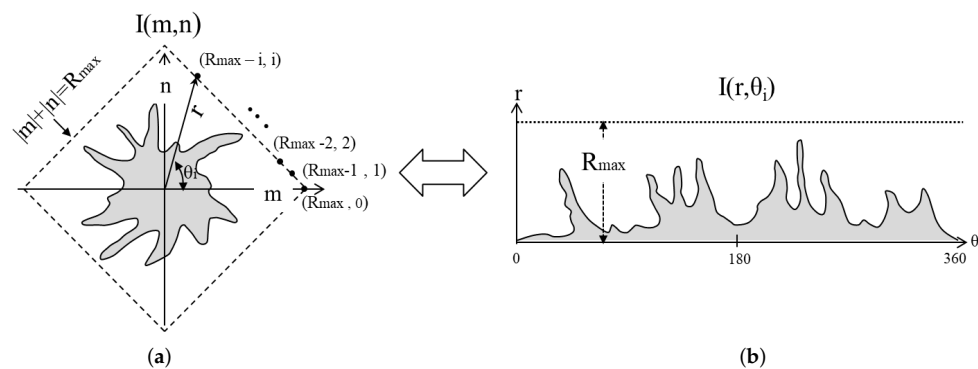
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-  $R_{max}$ : Pyramid radius enclosing the image.
-  $I(m, n)$ : Pixel value at  $(m, n)$  in the Cartesian coordinate system.
-  $I(r, i)$ : Pixel value at  $(r, i)$  in the discrete polar coordinate system.
-  $\theta_i$ : Legitimate angle of the  $i^{th}$  lattice on the pyramid of radius  $R_{max}$ .
/*Transform Cartesian coordinate to discrete polar coordinate. */
For  $i = 0$  to  $(4 \cdot R_{max} - 1)$  /* Determine legitimate angles. */
     $\theta = \tan^{-1} \left[ \frac{\text{mod}(i, R_{max})}{R_{max} - \text{mod}(i, R_{max})} \right]$  and then
     $\theta_i = \begin{cases} \theta & \text{for } i = 0, \dots, (R_{max} - 1) \\ 180^\circ - \theta & \text{for } i = R_{max}, \dots, (2R_{max} - 1) \\ 180^\circ + \theta & \text{for } i = 2R_{max}, \dots, (3R_{max} - 1) \\ 270^\circ - \theta & \text{for } i = 3R_{max}, \dots, (4R_{max} - 1) \end{cases}$ 
For  $r = 0$  to  $R_{max}$  /* Convert the coordinates lying at  $\theta_i$ . */
     $|n| = \partial \left( \frac{r \cdot |\tan \theta|}{1 + |\tan \theta|} \right)$  and  $(m, n) = \begin{cases} (r - |n|, |n|) & \text{for } 0^\circ \leq \theta_i < 90^\circ \\ (-r + |n|, |n|) & \text{for } 90^\circ \leq \theta_i < 180^\circ \\ (-r + |n|, -|n|) & \text{for } 180^\circ \leq \theta_i < 270^\circ \\ (r - |n|, -|n|) & \text{for } 270^\circ \leq \theta_i < 360^\circ \end{cases}$ 
     $I(r, i) \leftarrow I(m, n)$  /* Converting pixel values */
End /* of  $r$  */
End /* of  $i$  */
/*Transform polar coordinate to Cartesian coordinate. */
For  $i = 0$  to  $(4 \cdot R_{max} - 1)$ 
    For  $r = 0$  to  $R_{max}$  /* Convert the coordinates lying at  $\theta_i$  */
         $m = \partial(r \cos[\theta_i]), \quad n = \partial(r \sin[\theta_i])$ 
         $I(m, n) \leftarrow I(r, i)$ . /* Converting pixel values */
    End /* of  $r$  */
End /* of  $i$  */

```

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**Figure 7.** Image conversion between Cartesian and polar coordinates. (a) Image in the discrete Cartesian coordinate. (b) Image in the discrete polar coordinate.

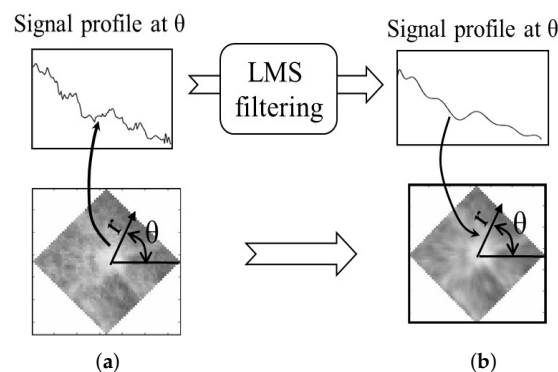
## 5. Applications of Radial Image Processing

This section discusses several applications of the proposed radial image processing method.

### 5.1. Smoothing Radially Oriented Images

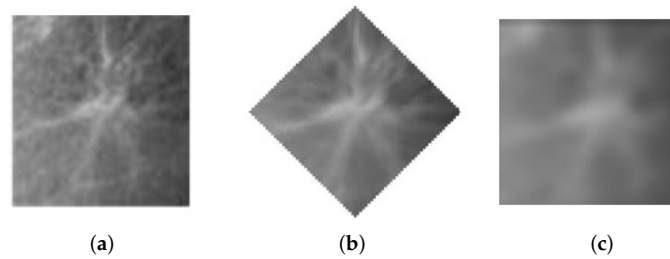
Consider an image of a potentially cancerous mass of breast tissue where the masses diffuse outward from its center radially. Since the local irregularities typically appear on such images, for an efficient processing of the images, it is necessary to smoothen the local irregularities while maintaining the radial directional properties of the images [2].

Figure 8 depicts the proposed radial smoothing method. Since the main signal flow of the image goes along the radial direction in the polar coordinate, the proposed method applies the least mean squared (LMS) filtering to the pixels at the same angle. LMS filtering reduces local irregularities in the best match with the radial characteristics of pixel signals [13]. The smoothing process is completed by performing the LMS processing at all angles.



**Figure 8.** Procedure of the proposed radial smoothing method. (a) Original mass image; (b) smoothed image.

Figure 9 shows a comparison between the original image, an image smoothed using Gaussian filtering [14], and an image smoothed using the proposed radial filter, respectively. The filter order was set to 7 for both smoothing filters for comparative purposes. As observed from Figure 9, the Gaussian filter blurs the overall shape of the mass since it neglects the directions of image signals and processes the image horizontally and vertically. However, the proposed method reduces the local irregularities of the image signal while maintaining the overall radial characteristics of the mass image. The proposed method also stabilizes the image optimally since the LMS filter adaptively smoothen signals at each direction.



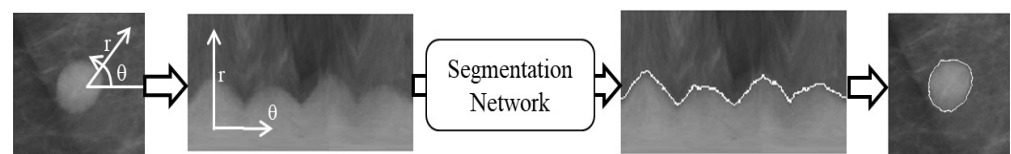
**Figure 9.** Comparison of smoothed mass images: (a) original mass image, (b) image smoothed by the proposed method, and (c) image smoothed by Gaussian filtering.

## 5.2. Biomedical Object Segmentation with Learning Network

Many biomedical objects tend to diffuse radially outward from the center. Such objects are often locally irregular in pixel values and their boundaries are locally ambiguous, making it difficult to segment and analyze the shapes of the objects. Segmenting such objects is useful for scientific analysis. For example, the analysis of mass contours in mammographic images is valuable for the detection of mass-type breast cancer [3,15]. Figure 7 illustrates a typical spiculated cancer in the Cartesian and discrete polar coordinate systems. In the Cartesian coordinate, the cancer edge gradients deviate from horizontal and vertical directions and are much variant, causing the shapes of these objects to be deformable and variant. In the polar coordinate, the edge gradients are directed along either the radius or angle direction, indicating that the boundary directions are less variant and the object shapes are more homogeneous.

It is well-known that learning-based methods are efficient in inferring ambiguous boundaries of biomedical objects. The learning performances deteriorate for images that are non-homogeneous with the training dataset images. Thus, in the Cartesian coordinate, severe edge variations of biomedical objects degrade the segmentation performances of learning networks. However, since the boundaries of biomedical objects are less variant in the polar coordinate system as discussed in Section 4, a learning network more efficiently recognizes the local edge gradients in the polar coordinate, thereby better tracking local edges and bridging the ambiguous edges in the polar coordinate.

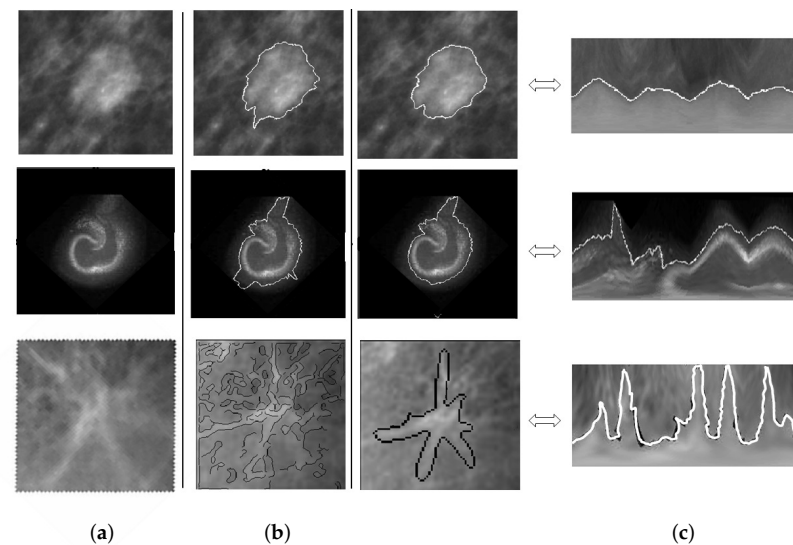
We transformed the biomedical images to the images in the polar coordinate and used Mask-RCNN to segment the biomedical objects [9,16,17]. The backbone of the MASK-RCNN was Resnet-50, and the learning rate parameter was set as 0.00025. Figure 10 illustrates the procedure of the proposed method. The training data set of approximately 1000 images was constructed with 700 images from the Pascal VOC dataset [18,19] and 300 images from our medical and biocell dataset. The training images selected from the Pascal dataset were chosen to be similar to deformable objects. The dataset was increased three times through data augmentation with flip, shift, and rotation. In the training phase, the dataset was split into training and testing data with a proportion of 80 to 20.



**Figure 10.** Illustration of the proposed biomedical object segmentation method.

Figure 11 shows the segmentation of a benign tumor in a mammographic image, a hippocampus in a microscopic biocell image, and the spiculate-shaped cancer in a mammographic image. As seen in the figure, the segmentation performances are superior when images in the Cartesian coordinate are transformed to those in the polar coordinate. Notably, the spiculated object that is extremely irregular in the Cartesian coordinate is rarely segmented in the Cartesian coordinate, whereas it is reasonably segmented in the polar coordinate. As explained in Figure 7, the spiculated object transformed in the polar

coordinate is much less variant, enabling the learning network to easily recognize the boundary of the object.



**Figure 11.** Segmentation results of biomedical objects. (a) Bio-cell images. From the top: benign tumor, hippocampus, and spiculated cancer. (b) Segmentation in Cartesian coordinate; (c) proposed segmentation in polar coordinate.

In terms of an ablation study, we compared and analyzed the results of direct segmentation in the Cartesian coordinates with those of segmentation in the polar coordinates by applying the proposed method. Table 1 compares the segmentation performances. To measure the effectiveness of segmentation, metrics including precision, recall, and F-Measure were used [20,21]. The performance of segmentation in the polar coordinate outperformed that in the Cartesian coordinate. Segmenting the spiculated cancer was the most difficult because the shape is mostly irregular and deformable. As seen in the table, the effect of segmentation in polar coordinates is the most tapped for the spiculated cancer, indicating that the proposed method is more efficient for more irregular and deformable biomedical objects.

**Table 1.** Deformable object segmentation performances of Cartesian and polar coordinates.

|                  | Recall    |       | Precision |       | Fscore    |       |
|------------------|-----------|-------|-----------|-------|-----------|-------|
|                  | Catersian | Polar | Cartesian | Polar | Cartesian | Polar |
| Benign tumor     | 0.87      | 0.96  | 0.86      | 0.97  | 0.81      | 0.97  |
| Hippocampus      | 0.70      | 0.85  | 0.84      | 0.97  | 0.74      | 0.90  |
| Spiculate cancer | 0.50      | 0.72  | 0.20      | 0.81  | 0.47      | 0.74  |

## 6. Conclusions and Further Study

In this paper, we propose a method that processes radial directional images, such as biomedical images, in  $L^1$ -norm-based discrete polar coordinates. We address the fact that only  $L^1$ -norm-based polar coordinates can exist in the discrete image domain and develop a discrete coordinate transformation algorithm between Cartesian coordinates and the  $L^1$ -norm-based polar coordinates. The transformation method perfectly converts between coordinates without causing any transform mismatches. The proposed radial method greatly reduces the directional variance of the radial directional images in the Cartesian coordinate system. The proposed method preserves the radial characteristics of such images, whereas processing in the Cartesian coordinate disperses the radial characteristics of the underlying images.

Since the polar coordinate processing sets the center of an image as the origin, the results of the proposed method are critically affected by how the center of the image is determined. Therefore, further studies are needed to determine the center of the image to maximize the performance of the proposed method.

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